Math 1610, Honors Calculus, Fall 2026, practice problems for Exam 2

Some definitions and results you may want.

Let f be defined on an open interval containing c. We say f is **differentiable at** c if the limit

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

exists. We say c is a **local maximum** (**local minimum**) for f if there is an $h > 0$ with $f(c) \geq f(x)$ (respectively $f(c) \leq f(x)$) for all $x \in (c-h, c+h)$ in the domain of f.

Sign preservation. If g is continuous at c and $g(c) > 0$, then there is an $h > 0$ with $g(x) > 0$ for every $x \in (c-h, c+h)$.

Some theorems from lecture. If f is continuous on $[a, b]$, then f takes every value between $f(a)$ and $f(b)$ (intermediate value theorem); f attains a maximum and a minimum value on $[a, b]$ (extreme value theorem); and f is uniformly continuous on $[a, b]$. If in addition f is differentiable on (a, b), then $f(b) - f(a) = f'(c)(b-a)$ for some $c \in (a, b)$ (mean value theorem). If f is continuous at c, then $F(x) = \int_a^x f(t)\, dt$ is differentiable at c with $F'(c) = f(c)$ (fundamental theorem of calculus).

Cauchy–Schwarz. For $\vec{a}, \vec{b} \in \mathbb{R}^n$, with $\vec{a} \cdot \vec{b} = \sum_i a_i b_i$ and $\|\vec{a}\| = \sqrt{\vec{a} \cdot \vec{a}}$,

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\, \|\vec{b}\|,$$

with equality if and only if one of $\vec{a}, \vec{b}$ is a scalar multiple of the other.

1. Show, directly from the definition of the limit, that

$$\lim_{x \to 2} \left(x^2 + x\right) = 6.$$

(Write $x^2 + x - 6 = (x-2)(x+3)$, and first restrict to $\delta \leq 1$ to get a bound on $|x+3|$.)

2. Prove the sign preservation lemma stated above.

3. Let $f : [a, b] \to \mathbb{R}$ be continuous, with $f(x) > 0$ for every $x \in [a, b]$.
   (a) Show that there is a number $m > 0$ with $f(x) \geq m$ for every $x \in [a, b]$.

   (b) Give an example showing that this can fail if $[a, b]$ is replaced by the interval $[0, \infty)$.

4. Let $f : [0, 2] \to \mathbb{R}$ be continuous, with $f(0) = f(2)$. Show that there is some $x \in [0, 1]$ with

$$f(x) = f(x+1).$$

(Consider $g(x) = f(x+1) - f(x)$ on $[0, 1]$, and compare $g(0)$ with $g(1)$.)

5. (Negating uniform continuity.)

   (a) Write out, without using the word "not", what it means for $f : S \to \mathbb{R}$ to fail to be uniformly continuous on S.

   (b) Show that $f(x) = 1/x$ is not uniformly continuous on the open interval $(0, 1)$.
   (Compare f at $\frac{1}{n}$ and at $\frac{1}{2n}$.)

6. Let $f(x) = x\,|x|$.

   (a) Show, from the definition, that f is differentiable at 0 with $f'(0) = 0$.

   (b) Compute $f'(x)$ for $x \neq 0$, and write down the function f' on all of $\mathbb{R}$.

   (c) Is f' differentiable at 0?

7. (Symmetric difference quotients.)

   (a) Suppose f is differentiable at c. Show that

$$\lim_{h \to 0} \frac{f(c+h) - f(c-h)}{2h} = f'(c).$$

   (b) Show that for $f(x) = |x|$ the limit in part (a) exists at $c = 0$, even though f is not differentiable there. So the converse of (a) is false.

8. Use the mean value theorem to show that for $0 < a < b$,

$$3a^2(b-a) \;<\; b^3 - a^3 \;<\; 3b^2(b-a).$$

9. Let f be continuous on $[a, b]$ and twice differentiable on (a, b), with $f(a) = f(b) = 0$, and suppose $f(c) > 0$ for some $c \in (a, b)$.

   (a) Use the mean value theorem on $[a, c]$ and on $[c, b]$ to find points $\xi_1 < \xi_2$ with $f'(\xi_1) > 0$ and $f'(\xi_2) < 0$.

   (b) Show that $f''(\xi) < 0$ for some $\xi \in (a, b)$.

10. (The second derivative test, without continuity of f''.) In lecture we proved the second derivative test assuming that f'' is continuous at the critical point. This problem shows that the hypothesis can be dropped.

    Let f be differentiable on an open interval (a, b), let $c \in (a, b)$ with $f'(c) = 0$, and suppose that $f''(c)$ exists and $f''(c) > 0$. (We do not assume f'' exists anywhere else, let alone that it is continuous.)

    (a) Define

$$Q(x) = \begin{cases} \dfrac{f'(x) - f'(c)}{x - c} & x \neq c, \\ f''(c) & x = c. \end{cases}$$

    Show that Q is continuous at c.

    (b) Show that there is an $h > 0$ such that $f'(x) < 0$ for $x \in (c - h, c)$ and $f'(x) > 0$ for $x \in (c, c + h)$.

    (c) Conclude that f has a local minimum at c.

11. Let $f(x) = x^3 - 3x$.

(a) Find the critical points of f, and use the second derivative test to decide whether each is a local maximum or a local minimum.

(b) Explain why f attains a largest and a smallest value on $[-2, 3]$, and why each must occur either at an endpoint or at a critical point. Find both values, and every point where they are attained.

(c) How many solutions does the equation $x^3 - 3x = 1$ have? Prove your answer.

12. (The fundamental theorem.) In parts (b) and (c), f is continuous on $\mathbb{R}$.

(a) Using the evaluation rule (the second fundamental theorem of calculus), compute

$$\int_1^2 \left(3x^2 - \frac{2}{x^2}\right) dx,$$

and check that the antiderivative you use really has the integrand as its derivative.

(b) Find a continuous f with $\int_0^x f(t)\,dt = x^3 + x$ for every x, and explain why it is the only one.

(c) Is there a continuous f with $\int_0^x f(t)\,dt = x^3 + 1$ for every x?

13. (Cauchy–Schwarz at work.)

(a) Show that for all positive real numbers a, b, c,

$$(a+b+c)\Big(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\Big) \geq 9.$$

(b) When does equality hold?

(Apply Cauchy–Schwarz to a well-chosen pair of vectors in $\mathbb{R}^3$.)

14. (Vector-valued functions.)

(a) Show, directly from the definition, that $\lim_{t\to 0} (t,\, t^2) = (0, 0)$.

(b) Give an example of a function $\vec{F} : (-1, 1) \to \mathbb{R}^2$ such that $\lim_{t\to 0} \|\vec{F}(t)\|$ exists but $\lim_{t\to 0} \vec{F}(t)$ does not. Justify both claims.

(c) Let $\vec{F}(t) = \big(t^3,\, \frac{1}{t}\big)$ for $t > 0$. Compute $\vec{F}\,'(t)$ and $\vec{F}\,'(1)$, and use $\vec{F}(1+h) \approx \vec{F}(1) + h\,\vec{F}\,'(1)$ to estimate $\vec{F}(1.1)$. Compare with the true value.